

Laplace Transform Properties — Full Derivation Cheat Sheet

(Companion document to the Laplace Transform Pairs cheat sheet. All signals are causal: $x(t) = 0$ for $t < 0$.)

These 13 properties are the actual **tools** used to build every transform pair in the companion table — this document derives the tools themselves, from the definition alone:

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt$$

1. Multiplication by Constant

$$K x(t) \iff K X(s)$$

Derivation. The integral operator is linear, so constants pull straight out:

$$\mathcal{L}\{Kx(t)\} = \int_0^{\infty} Kx(t)e^{-st} dt = K \int_0^{\infty} x(t)e^{-st} dt = \boxed{KX(s)}$$

2. Linearity

$$K_1 x_1(t) + K_2 x_2(t) \iff K_1 X_1(s) + K_2 X_2(s)$$

Derivation. Same idea, extended to a sum term-by-term (splitting the integral of a sum into a sum of integrals):

$$\mathcal{L}\{K_1 x_1 + K_2 x_2\} = \int_0^{\infty} [K_1 x_1(t) + K_2 x_2(t)] e^{-st} dt = K_1 \int_0^{\infty} x_1 e^{-st} dt + K_2 \int_0^{\infty} x_2 e^{-st} dt = \boxed{K_1 X_1(s) + K_2 X_2(s)}$$

3. Time Scaling

$$x(at), a > 0 \iff \frac{1}{a} X\left(\frac{s}{a}\right)$$

Step 1 — substitute $\tau = at$, so $t = \tau/a$ and $dt = d\tau/a$. Since $a > 0$, the limits $t:0 \rightarrow \infty$ map to $\tau:0 \rightarrow \infty$ (no flip):

$$\mathcal{L}\{x(at)\} = \int_0^{\infty} x(at) e^{-st} dt = \int_0^{\infty} x(\tau) e^{-s\tau/a} \frac{d\tau}{a}$$

Step 2 — pull out the constant $1/a$ and recognize the remaining integral as $X(\cdot)$ evaluated at s/a :

$$= \frac{1}{a} \int_0^{\infty} x(\tau) e^{-(s/a)\tau} d\tau = \boxed{\frac{1}{a} X\left(\frac{s}{a}\right)}$$

4. Time Shift

$$x(t - T)u(t - T) \iff e^{-Ts}X(s)$$

Step 1 — since $u(t - T) = 0$ for $t < T$, the lower limit becomes T :

$$\mathcal{L}\{x(t - T)u(t - T)\} = \int_T^{\infty} x(t - T)e^{-st} dt$$

Step 2 — substitute $\tau = t - T$ (limits shift from $T \rightarrow \infty$ down to $0 \rightarrow \infty$):

$$= \int_0^{\infty} x(\tau) e^{-s(\tau+T)} d\tau = e^{-Ts} \int_0^{\infty} x(\tau) e^{-s\tau} d\tau = \boxed{e^{-Ts}X(s)}$$

5. Frequency Shift

$$e^{-at}x(t) \iff X(s + a)$$

Derivation. Absorb the exponential directly into the transform kernel:

$$\mathcal{L}\{e^{-at}x(t)\} = \int_0^{\infty} x(t) e^{-at} e^{-st} dt = \int_0^{\infty} x(t) e^{-(s+a)t} dt$$

This is precisely the definition of $X(\cdot)$ evaluated at $(s + a)$ instead of s :

$$\boxed{X(s + a)}$$

6. Time 1st Derivative

$$x'(t) = \frac{dx}{dt} \iff sX(s) - x(0^-)$$

Step 1 — integrate by parts, with $u = e^{-st}$ ($du = -se^{-st}dt$) and $dv = x'(t)dt$ ($v = x(t)$):

$$\mathcal{L}\{x'(t)\} = \int_0^\infty x'(t)e^{-st} dt = [x(t)e^{-st}]_0^\infty - \int_0^\infty x(t)(-s)e^{-st} dt$$

Step 2 — evaluate the boundary term (assuming $x(t)e^{-st} \rightarrow 0$ as $t \rightarrow \infty$, which holds wherever $X(s)$ converges):

$$[x(t)e^{-st}]_0^\infty = 0 - x(0^-) = -x(0^-)$$

Step 3 — combine:

$$\mathcal{L}\{x'(t)\} = -x(0^-) + s \int_0^\infty x(t)e^{-st} dt = \boxed{sX(s) - x(0^-)}$$

7. Time 2nd Derivative

$$x''(t) = \frac{d^2x}{dt^2} \iff s^2X(s) - sx(0^-) - x'(0^-)$$

Derivation. Apply Property 6 twice. Let $y(t) = x'(t)$, so $Y(s) = sX(s) - x(0^-)$ by Property 6. Apply Property 6 again, now to y :

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0^-)$$

Substitute $Y(s) = sX(s) - x(0^-)$ and $y(0^-) = x'(0^-)$:

$$\mathcal{L}\{x''(t)\} = s[sX(s) - x(0^-)] - x'(0^-) = \boxed{s^2X(s) - sx(0^-) - x'(0^-)}$$

(This pattern continues: each additional derivative multiplies by another s and subtracts one more initial-condition term.)

8. Time Integral

$$\int_{0^-}^t x(\tau) d\tau \iff \frac{1}{s}X(s)$$

Step 1 — define $g(t) = \int_{0^-}^t x(\tau) d\tau$. By the Fundamental Theorem of Calculus, $g'(t) = x(t)$, and $g(0^-) = 0$ (empty integral).

Step 2 — apply Property 6 to $g(t)$ instead of $x(t)$:

$$\mathcal{L}\{g'(t)\} = sG(s) - g(0^-)$$

Step 3 — substitute $\mathcal{L}\{g'(t)\} = \mathcal{L}\{x(t)\} = X(s)$ and $g(0^-) = 0$, then solve for $G(s)$:

$$X(s) = sG(s) - 0 \Rightarrow \boxed{G(s) = \frac{X(s)}{s}}$$

9. Frequency Derivative

$$t x(t) \iff -\frac{d}{ds} X(s) = -X'(s)$$

Derivation. Differentiate the *definition itself* with respect to s , moving the derivative inside the integral:

$$\frac{dX}{ds} = \frac{d}{ds} \int_0^\infty x(t) e^{-st} dt = \int_0^\infty x(t) \cdot \frac{\partial}{\partial s} (e^{-st}) dt = \int_0^\infty x(t) (-t) e^{-st} dt$$

Recognize the right side as $-\mathcal{L}\{t x(t)\}$:

$$X'(s) = -\mathcal{L}\{t x(t)\} \Rightarrow \boxed{\mathcal{L}\{t x(t)\} = -X'(s)}$$

10. Frequency Integral

$$\frac{x(t)}{t} \iff \int_s^\infty X(q) dq$$

Step 1 — let $g(t) = x(t)/t$, so $x(t) = t g(t)$.

Step 2 — apply Property 9 to $g(t)$: $\mathcal{L}\{t g(t)\} = -G'(s)$. Since $\mathcal{L}\{t g(t)\} = \mathcal{L}\{x(t)\} = X(s)$:

$$X(s) = -G'(s)$$

Step 3 — integrate both sides with respect to s , from s up to ∞ :

$$\int_s^\infty X(q) dq = - \int_s^\infty G'(q) dq = -[G(\infty) - G(s)]$$

Step 4 — since $G(\infty) = 0$ (a convergent Laplace transform vanishes as $s \rightarrow \infty$):

$$\int_s^\infty X(q) dq = G(s) \Rightarrow \boxed{G(s) = \int_s^\infty X(q) dq}$$

11. Initial Value Theorem

$$x(0) = \lim_{s \rightarrow \infty} sX(s)$$

Step 1 — start from Property 6 in integral form:

$$sX(s) - x(0^-) = \int_0^\infty x'(t)e^{-st} dt$$

Step 2 — take $s \rightarrow \infty$ on both sides. For every fixed $t > 0$, $e^{-st} \rightarrow 0$, so the integral vanishes (assuming $x'(t)$ is well-behaved enough for the limit to pass inside):

$$\lim_{s \rightarrow \infty} [sX(s) - x(0^-)] = 0$$

Step 3 — solve for the limit:

$$\lim_{s \rightarrow \infty} sX(s) = x(0)$$

12. Final Value Theorem

$$\lim_{t \rightarrow \infty} x(t) = x(\infty) = \lim_{s \rightarrow 0} sX(s)$$

Step 1 — same starting identity as Property 11:

$$sX(s) - x(0^-) = \int_0^\infty x'(t)e^{-st} dt$$

Step 2 — take $s \rightarrow 0$ instead. Now $e^{-st} \rightarrow 1$, so the right side becomes an ordinary definite integral of $x'(t)$:

$$\lim_{s \rightarrow 0} [sX(s) - x(0^-)] = \int_0^\infty x'(t) dt = [x(t)]_0^\infty = x(\infty) - x(0^-)$$

Step 3 — the $x(0^-)$ terms cancel from both sides:

$$\lim_{s \rightarrow 0} sX(s) = x(\infty)$$

(Valid only when $x(\infty)$ actually exists — i.e., all poles of $sX(s)$ lie in the open left half-plane, aside from a single pole allowed at $s = 0$.)

13. Convolution

$$x_1(t) * x_2(t) \iff X_1(s)X_2(s)$$

Step 1 — write out the (causal) convolution and its transform:

$$y(t) = x_1(t) * x_2(t) = \int_0^t x_1(\tau)x_2(t-\tau) d\tau, \quad Y(s) = \int_0^\infty y(t)e^{-st} dt$$

Step 2 — substitute and swap the order of integration. The region $0 \leq \tau \leq t < \infty$ can equivalently be described as τ from 0 to ∞ , with t from τ to ∞ :

$$Y(s) = \int_0^\infty \int_0^t x_1(\tau)x_2(t-\tau)e^{-st} d\tau dt = \int_0^\infty x_1(\tau) \left[\int_\tau^\infty x_2(t-\tau)e^{-st} dt \right] d\tau$$

Step 3 — in the inner integral, substitute $u = t - \tau$ ($du = dt$, limits $0 \rightarrow \infty$):

$$\int_\tau^\infty x_2(t-\tau)e^{-st} dt = \int_0^\infty x_2(u)e^{-s(u+\tau)} du = e^{-s\tau} \int_0^\infty x_2(u)e^{-su} du = e^{-s\tau} X_2(s)$$

Step 4 — substitute back and factor out the constant $X_2(s)$:

$$Y(s) = \int_0^\infty x_1(\tau) e^{-s\tau} X_2(s) d\tau = X_2(s) \int_0^\infty x_1(\tau) e^{-s\tau} d\tau = \boxed{X_1(s)X_2(s)}$$

Summary: The Dependency Chain

Property	Built from
1, 2	Definition (linearity of integration)
3, 4, 5	Definition (substitution / direct algebra)
6	Definition (integration by parts)
7	Property 6, applied twice
8	Property 6, applied to the antiderivative of $x(t)$
9	Definition (differentiate under the integral sign)
10	Property 9, then integrate the result
11, 12	Property 6, evaluated at the limits $s \rightarrow \infty$ and $s \rightarrow 0$
13	Definition (double integral + substitution)

The big picture: Properties 6 (derivative) is the single most load-bearing result in the table — Properties 7, 8, 11, and 12 all derive directly from it. Properties 9 and 10 form their own small pair (differentiate/integrate in s , mirroring 6/8's differentiate/integrate in t). Once Property 6 and the definition itself are second nature, the rest of the table falls out with one or two lines of algebra each — exactly the same relationship the *transform pairs* cheat sheet has to *this* properties cheat sheet: a small kernel of ideas, reused everywhere.